OBD Config Planning and Requirement

Updates

- Date — 30/10/2026
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Problem Statement

Manually creating or editing a JSON configuration file can introduce syntax errors, missing fields, unsupported values, and incorrect combinations of settings. These errors may prevent the logger from operating as intended.

Goals and Objectives

- Provide an understandable interface for configuring logging.
- Generate config.json in the exact structure expected by the logger firmware
- Validate required fields, data types, ranges, and dependent settings
- Reduce the need for users to edit JSON manually.
- Allow users to review their settings before exporting.
- Provide clear feedback when a configuration is incomplete or invalid.

Scope

- Configuration entry and editing
- Validation against the supported configuration format.
- Generation and export of config.json
- Documentation describing configuration fields and supported values.
- Importing an existing configuration.
- Supporting multiple standards configuration versions — TBD

Requirements Gathering

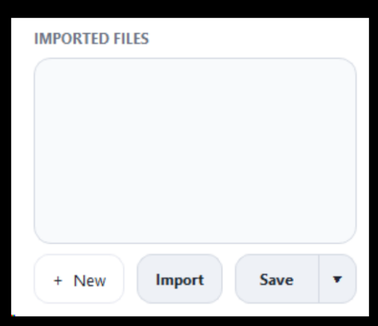
- exact JSON key
- Data type and unit
- Default value, if one exists
- Allowed values or minimum and maximum
- Firmware versions supporting it
- Expected error message for invalid input
- A valid example.

Requirements Analysis

- Configuration entry — Login info — TBD
- Required-field validation — The software shall identify missing required values before export.
- Error feedback — The software shall identify the affected field and explain how to correct an invalid value.
- Configuration review — The software shall allow users to review the configuration before export
- JSON generation — The software shall generate syntactically valid JSON using the exact keys, expected by the supported firmware.
- Invalid-export prevention — The software shall prevent export while known validation errors remain.
- Transfer guidance — The software shall explain where the exported file must be placed on the SD card, according to the logger’ s documented requirements.

UI-Features

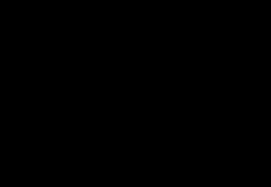
- Login — TBD
- Import —



- PID selection Based on J1979-DA —

PID	Comment
01	Monitor status since DTCs cleared
04	Calculated load value
05	Engine coolant temperature
0B	Intake manifold absolute pressure
0C	Engine RPM
0D	Vehicle speed sensor
0F	Intake air temperature
10	Air flow rate from mass air flow sen...
11	Absolute throttle position
13	Location of oxygen sensors
1C	OBD requirements to which vehicle ...
21	Distance traveled while MIL is activat.
23	Fuel rail pressure
24	Equivalence ratio
2C	Commanded EGR
2D	EGR error
30	Number of warm-ups since DTCs cl...
31	Distance traveled since DTCs cleared
33	Barometric pressure
41	Monitor status this driving cycle
42	Control module voltage
46	Ambient air temperature

- Themes and Help



- JSON Preview

